



arxiv.org/abs/2504.04715

Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs: Why Software Tests Fail and Hardware Attestation Wins. Commercial providers face strong economic incentives to covertly serve cheaper models — and current detection methods are fundamentally inadequate.

ML Researchers

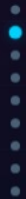
AI Security

Cloud Infrastructure

Analytical · Evidence-Driven · Technical

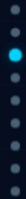
— THE PROBLEM

Model Substitution: **Economic Incentives** Drive Covert Swaps



— EMPIRICAL EVIDENCE

FP8 Quantization **Preserves Most Accuracy** — Making Substitution Hard to Detect



— TEMPERATURE ANALYSIS

Higher Temperature **Amplifies Performance Gaps** but Increases Noise



— FINGERPRINTING FAILURE

Log Probability Fingerprinting Fails Due to **Inference Nondeterminism**



— ADVERSARIAL COUNTERMEASURE

Randomized Routing at Low Rates Makes Statistical Detection **Query-Prohibitive**



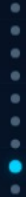
— FAILURE MODE ANALYSIS

Software-Only Auditing Methods Are Fundamentally Unreliable



— — THE SOLUTION

Trusted Execution Environments: Provable Cryptographic Model Integrity



— SUMMARY

Key Takeaways: Close the Verification Gap with Hardware Attestation

01

Substitution Is Economically Motivated

GPU costs create strong financial incentives for providers to covertly serve cheaper models under advertised pricing — and the black-box API gives users no recourse.

02

FP8 Quantization Is Nearly Undetectable

Benchmark drops are often within error bars — MMLU, GSM8K, and MATH scores are statistically indistinguishable across most model families, making benchmark audits futile.

03

Nondeterminism Kills Log-Prob Fingerprinting

Inference framework versions (transformers, vLLM) and GPU hardware (A100 vs H100) each produce distinguishable log-probability shifts — indistinguishable from substitution.

05

TEEs Are the Only Currently Viable Path to Verifiable LLM APIs

Hardware-secured enclaves (e.g., NVIDIA confidential computing on H100) provide cryptographic attestation that correct weights are loaded and correct code runs — provable, efficient, and robust against all adversarial countermeasures. The community needs to standardize attestation interfaces and audit tooling around TEE-enabled inference.

04

Randomized Routing Defeats Statistical Tests

At low substitution rates p , the mixed distribution converges to the specified model's distribution — requiring exponentially more queries, and enabling undetected partial substitution.